

## JWST Mirror Quest – Interactive Space Science Game

These is a Hackathon project

Learn by playing – discovering the universe, one hexagon at a time.

### 1. Main Problem and Goal

- The problem is that although NASA's James Webb Space Telescope (JWST) makes remarkable discoveries, its complicated engineering and the data it receives from space are still difficult for the average public to understand easily.
- Objective: To reduce the gap between space science and public engagement by means of an interactive and gamified educational app designed for all age groups.

### 2. Product Architecture and Key Features

- Mirror-Based Navigation System: The interface of the app directly reflects the 18 hexagonal mirror sections of the telescope.
- Nested Game Progression:

There are 18 main mirror levels, each primary segment corresponding to a fundamental scientific topic or capability.

324 Total Sub-Levels. Each main segment splits into 18 sub-levels or quizzes.

By using interactive features, completing the sub-levels causes the main mirror segments to be assembled and aligned, thereby revealing genuine NASA deep-space images.

- Adaptive Quiz Engine:
  - For incorrect answers, give hints that are dynamic and aware of the context.
  - For correct answers, you can access detailed scientific explanations together with high-resolution space photography.

### 3. Technical Architecture

- Frontend: React Native (it provides cross-platform compatibility with iOS, Android, and Web).
- Backend and database: A cloud database (Firebase) is used to manage the 324 mini-hex states, user progression, and the delivery of questions.
- Data Integration: NASA's open datasets and image APIs providing real space imagery.

### 4. Progress in technical status and implementation

- The design of the user interface and user experience features a main mirror grid layout and a nested 18-tile navigation structure that is now complete.
- Question Pipeline: Content database filled with questions, step-by-step hints, and detailed post-quiz breakdowns.
- Working Prototype: An interface that is fully functional and includes the ability to select the difficulty level (Easy/Medium/Hard), the quiz flow, and animations for segment alignment.

### 5. Value Proposition and Future Growth

- Educational Impact: It advances STEM education by converting complicated astronomical concepts into interactive visual rewards.
- Future Developments:

Available are leaderboards and the option to play trivia games online with other players.

Physical companion card games for classrooms.

We add new content right after live NASA mission discoveries.